EXPERIMENT NO: 1

STUDY OF VARIOUS COMMANDS USED IN LINUX AND WINDOWS

AIM

To study and practice various file handling and networking commands used in Linux and Windows operating systems.

BASIC LINUX COMMANDS

1. pwd

Description: Prints the current working directory

```
$ pwd
/home/student
```

2. mkdir

Description: Creates a new directory

```
$ mkdir netlab
$ ls
netlab
```

3. cd

Description: Changes the current directory

```
$ cd netlab
$ pwd
/home/student/netlab
```

4. touch filename.txt

Description: Creates an empty file

```
$ touch notes.txt
$ ls
notes.txt
```

5. echo "text" > filename.txt

Description: Writes text into a file

```
$ echo "Networks Lab" > notes.txt
```

6. cat filename.txt

Description: Displays the contents of a file

```
$ cat notes.txt
Networks Lab
```

7. grep

Description: Searches for a pattern in a file

```
$ grep "Lab" notes.txt
```

```
Networks Lab
```

8. wc

Description: Displays the number of lines, words and bytes in a file

```
$ wc notes.txt
1  2  14 notes.txt
```

9. wc -l

Description: Counts the number of lines in a file

```
$ wc -l notes.txt
1 notes.txt
```

10. wc -w

Description: Counts the number of words in a file

```
$ wc -w notes.txt
2 notes.txt
```

BASIC NETWORKING COMMANDS

1. hostname

Description: Displays the system's hostname

```
$ hostname
student-pc
```

2. ifconfig / ip address show

Description: Displays IP address and network interface details

```
$ ip address show
1: lo
2: eth0   inet 192.168.1.14/24
```

3. ping

Description: Tests connectivity to a host

```
$ ping -c 4 google.com
64 bytes from ...: time=12 ms
4 packets transmitted, 4 received, 0% loss
```

4. nslookup

Description: Finds the IP address of a domain

```
$ nslookup google.com
Name: google.com
Address: 142.250.193.78
```

5. arp -a

Description: Displays the ARP cache

```
$ arp -a
192.168.1.1  (gateway) at 3c:84:6a:7e:04:22
```

6. traceroute

Description: Displays the route taken by packets to a destination

```
$ traceroute google.com
1  gateway (192.168.1.1)  1.2 ms
2  10.10.0.1  5.4 ms
```

WINDOWS EQUIVALENT COMMANDS

ls → dir cat → type cp → copy rm → del traceroute → tracert

RESULT

The basic Linux and Windows networking commands were executed successfully and the expected outputs were obtained.